

UMAA MAHESHWARY SV

AI/ML & DEEP LEARNING ENGINEER | SOFTWARE & FULL-STACK AI DEVELOPMENT | DIGITAL & BUSINESS STRATEGY

India | +91 8618479206 | umaamaheshwarysv@gmail.com

LinkedIn: [umaa-maheshwary-sv](#) | GitHub: [Umaa-6183](#) | Portfolio: [portfolio-umaa.vercel.app](#)

PROFESSIONAL SUMMARY

Computer Science professional with hands-on experience in Machine Learning, Deep Learning, Generative AI, NLP, and Intelligent Systems, skilled in Python, PyTorch, TensorFlow, and full-stack development (React, FastAPI, Node.js). Proven track record building AI solutions, quantum ML frameworks, and cloud-deployed (AWS, Azure, GCP) systems across healthcare, wellness, security, and career-tech domains. Strong research, communication, and digital marketing skills suited to both technical and business-facing roles.

CORE COMPETENCIES

Machine Learning & AI Engineering | Applied AI & Generative AI | LLMs & Prompt Engineering | Full-Stack Web & Application Development | Software Development | AI Application Development | Cloud Computing & Deployment | Quantum Machine Learning | Computational Biology & Bioinformatics | Data Analysis & Visualization | Intelligent Systems & Multi-Agent Systems | Project & Product Management | Digital Marketing & SEO | Business & Technical Communication | Cross-Functional Collaboration | Stakeholder Management | Problem Solving | Team Leadership

TECHNICAL SKILLS

Machine Learning & Deep Learning: Machine Learning, Deep Learning, Supervised Learning, Transformers, CNN, RNN, GNN, Multimodal Learning, Multi-Agent Systems, Generative AI, LLMs, NLP

Frameworks, Libraries & Programming: PyTorch, TensorFlow, Scikit-learn, PyTorch Geometric, Pandas, NumPy, Python, C, SQL, Bash, R

Full-Stack, Cloud & Quantum Computing: React, FastAPI, Node.js, AWS (EC2, S3), Microsoft Azure, GCP, AI Deployment Pipelines, Qiskit, PennyLane

Computational Biology, Bioinformatics & Data Science: Gene Expression Analysis, Network Modeling, Biopython, Scanpy, DeepChem, RDKit, Data Preprocessing, Statistical Analysis, Hypothesis Testing, Model Evaluation, Hyperparameter Tuning

Document Automation & Digital Marketing: Node.js/docx Automated Reporting, XML Validation, Google SEO, Meta Marketing, Social Media & Content Marketing

PROFESSIONAL EXPERIENCE

Research and Development Analyst Intern | JM Analytics and Solutions Private Limited (Oct 2025 – Aug 2026)

- Designed and deployed a multi-agent AI wellness platform with NLP-based conversational agents and supervised personalization models to boost engagement.

- Deployed and evaluated ML models on AWS EC2/S3, applying hyperparameter tuning and model evaluation to improve prediction performance and deployment stability.

Quantum Machine Learning (QML) Engineer Intern | Ada Lovelace Software Private Limited (Mar 2, 2026 – Jun 18, 2026)

- Built and analyzed a Hybrid Quantum ML framework combining genetic algorithms (Qiskit) for DNA sequence classification.

- Benchmarked classical vs. quantum ML approaches through simulation, visualization, and comparative analysis.

PROJECTS

Cross-Cultural Multi-Modal Emotion Recognition System -

Multimodal deep learning (BERT, CNN, LSTM) for cross-cultural emotion classification via feature-level and late fusion.

AIShieldX – Enterprise AI Security, Governance & Trust Platform -

Enterprise AI security, governance, monitoring, and trust; 20-vertical product catalog on a pnpm/Turborepo monorepo.

CareerX AI – Intelligent Career Growth Platform -

AI-powered platform for personalized career recommendations, job matching, and career growth.

MindfulPath – AI-Powered Wellness Platform - React/FastAPI wellness app with NLP-driven assessments, meditation/yoga content, Razorpay/Stripe payments, and region-gated pricing.

Anemia AI Tracker – Non-Invasive Hemoglobin Screening -

Non-invasive anemia screening via conjunctival image analysis and a custom-built neural network for hemoglobin estimation.

ASRMAS – Autonomous Scientific Research Multi-Agent System -

Multi-agent AI system (FastAPI, React, PennyLane) for autonomous scientific research and computational biology.

RESEARCH & PUBLICATION

Cross-Cultural And Context-Aware Multi-Modal Emotion Recognition System Using Deep Supervised Learning - Multimodal (BERT, CNN, LSTM) emotion classification with feature-level and late fusion; evaluated via Precision/Recall/F1, outperforming unimodal baselines. Published in a peer-reviewed UGC-approved journal.

DOI: [10.56726/IJRMETS82067](#)

CCDT: Cognitive Digital Twin for Cloud Operating System - Cognitive Digital Twin framework combining ML, predictive modeling, and cloud resource optimization for performance forecasting. Published in IJRAR, Vol. 13, Issue 2, May 2026, pp. 399–405 (Paper ID: IJRAR26B2422).

DOI: [10.56975/ijrar.v13i2.334330](#)

EDUCATION

M.Tech in Computer Science and Engineering | Rajarajeshwari College of Engineering

2026 | CGPA: 9.36 / 10

B.E. in Computer Science and Engineering | City Engineering College

2024 | CGPA: 6.58 / 10

CERTIFICATIONS & PROFESSIONAL TRAINING

Certifications: TensorFlow Developer Certificate (Google), AWS Certified Machine Learning - Specialty (AWS).

Courses & Training: Bioinformatics (Coursera), Deep Learning Specialization (DeepLearning.AI/Coursera), Multi-Omics (EMBL-EBI), Software Engineering Virtual Experience (JPMorgan Chase/Forage).